

Der Trog

Prototype for an embodied contactless Breath-To-Water-Interface

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Figure 1: Der Trog. Water surface detail.

Abstract

Der Trog is a prototype for a contactless, embodied breath-to-water interface, exploring the potential of Artificial Intelligence as a mediator to facilitate emotional and sensory interactions with nature. By tracking respiration via thermal image analysis and translating it into wave patterns on a water surface, the installation provides a tangible experience translating the user's corporal rhythm onto a natural material. The present research project investigates whether such interactions can help bridging the gap between environmental knowledge and action within the ecological crisis, encouraging holistic empathy for nature. Der Trog aims to inspire new perceptions of our connection to natural elements, advocating for sustainable behaviour through meaningful, embodied encounters with the environment.

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CCS Concepts

• **Human-centered computing** → **Interaction design**.

Keywords

Environmental Philosophy, Sustainability, Embodied Interaction, Computer Vision, Tangible Media

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1 Conceptual Background

1.1 From 'Environmental Knowledge' to 'Environmental Being'

The question to which extent Artificial Intelligence (AI) can help to manage the advancing global ecological crisis is currently focused on the rapid processing of large amounts of data to optimize the management of ecosystems. However, several voices in contemporary psychology, philosophy, design- and climate-research claim that *physical* interactions between ourselves and nature are needed to strengthen our self-awareness within our surroundings

[7, 21, 25], and ultimately return to a more sustainable way of living; experiences that recreate empathy for nature and our position therein [8, 11]. Accordingly, *Der Trog* addresses the question of if and how AI could be used as a mediator to generate new perceptions of our self in natural surroundings, by facilitating novel modes of interaction.

Dr. Joachim Rathmann, ecology-philosopher at the University of Augsburg in Germany, illuminates the ‘paradox of environmental knowledge’ in his 2023 essay ‘Artificial Intelligence and the Environment’ [26]. Despite long-standing global awareness and information needed for comprehensive nature and climate protection, negative trends in global ecosystem indicators persist [27], with limited real-world action to transition from overconsumption-driven lifestyles to more sustainable ones. In the ecological context, the power of images and the educational effect of rational information seems to be limited. Factual insights seem to hardly affect humans in a holistic and emotional way [10, 16, 20]. According to Rathmann, it is not information in a sense of environmental knowledge or awareness that is lacking, but rather awareness and an empathic connection in a sense of ‘environmental being’ [2]. In this definition, *holistic* empathy includes an embodied component [7]. He underscores the limitations of AI, noting its inability to address the emotional, ethical, and existential dimensions of environmental stewardship. While AI can determine the ‘what’ of ecological action, it lacks the capacity to answer the critical ‘why’ that motivates sustainable behavior.

Rathmann contends that the ecological crisis is a spiritual one, with humans in the Anthropocene becoming disconnected not only from themselves but also from nature, seeing it solely as an object and resource, devoid of the physical and spiritual vitality that makes it a counterpart. To overcome this crisis, he calls for an emotional and empathic bond with nature – an understanding of the environment (‘Umwelt’) as a ‘co-environment’ (‘Mitwelt’). Central to cultivating this empathy are ‘bodily’ encounters with nature and natural material, which could revive a corporal orientation towards nature as an intrinsically valuable experience that has to be protected [7]. He finally concludes that mindful, empathic interactions with our bodies and natural material can foster individual sensitivity and offer a motion contrary to the Western concept of economic progress and natural resource exploitation. Rathmann thus provides philosophical, yet practical advice on how to achieve the necessary rethinking to reach global climate goals, which stand and fall with the environmental behaviour of the single individuals within our society. However, Rathmann’s essay leaves a pivotal question unanswered: How can we interfere with the awareness and corporality of the self in interplay with our natural environment? *Der Trog* explores answers to this question by using AI as a mediator to generate new perceptions of our self in natural surroundings.

1.2 Redrawing Perception through Sensual, Embodied Experiences

Breathing, as a mostly unconscious yet influenceable process, not only connects us to our body and as such is essential for survival, it also ties us directly to our environment through the air we inhale. This is happening on multiple levels: On a micro level, breathing

builds a bridge to the immediately surrounding nature that supplies oxygen, as well as to other organisms whose exhaled air we breathe. On a macro level, the air we breathe represents the only connection from our earth to space. As the ‘dominating’ species on earth, however, the air we breathe is not only an element that surrounds us, it is also a manifestation of ourselves. The polluted air of the industrial world is a reflection of our way of life. Breathing therefore serves as a metaphor, a mirror of our relationship to nature, to ourselves, and to technological advancement. We breathe, sometimes consciously, sometimes unconsciously, but rarely aware of the profoundness of this seemingly banal process.

But how does awareness for an intangible dimension arise? How does the breath suddenly become an action that I consciously control with my muscles, with which I consciously ‘breathe life into myself’ and connect to my surroundings? Awareness arises through senses, through feeling, through confrontation: On a cold winter morning, I can see the air; I can blow small ‘steam sculptures’ and become aware that this is the same air I just breathed in, sparking fascination. Blowing on the tea in my hand to cool it, I’m reminded that this same dynamic also creates waves in the ocean. I have established an emotional, associative link through the experience of sensual feedback. This connection to my body in the context of my surroundings can evoke further reflection on larger constellations. Changing our perception means directing our attention, our emotional focus, to something that is yet unknown to us. A train of thought, a sensory experience, a feeling. To take a perspective that used to be hidden from us previously.

We can redraw the boundaries of perception by being receptive to new perspectives; concretely, new perspectives must reach us emotionally and on a holistic mind and body level. Within the context of environmental awareness and corporality, Artificial Intelligence has been ascribed the role of a cold technological entity, a mere numerical tool to present us with the factual knowledge we need to act and yet ignore, thus devoid of the ability to spark empathy and environmental engagement. We want to challenge this perspective: Could AI in fact take on a mediating role in this setting, an access-support to open us new, fascinating, emotional, and, above all, conscious perspectives to our surroundings, which ultimately influence our perception beyond the present moment?

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2.1 Design Considerations

If we understand AI as a tool to augment conventional physical interaction, it can open new ways for humans to interact with natural matter. Computer vision systems allow us to track and parametrize bodily processes such as our breath with non-invasive technology - cameras - and translate this process into an experiential work. The occurring, embodied interaction creates a direct and intuitive experience that gives the user a more natural sense of connection and interaction [5, 17]. If we further acknowledge that especially physical, sensory interaction creates conscious, embodied access [6, 7], we can think of creating tangible experiences using ‘natural’ materials and physical media by means of AI.

We tried to imagine unprecedented interactions that could make the user experience natural matter as a literal extension of their body. What if the trees in the forest moved back and forth to the



Figure 2: Installation view.

rhythm of my heartbeat? What if the river I am walking along flowed according to my pace? What if I could breathe the waves onto the ocean as I stand on the shore?

The constellation of water and (vibration) waves, as a cymatic process, appeared interesting to us, since it has been investigated in several earlier works. Most of these works, however, are either of performative manner – like Lisa Park’s ‘Eunoia II’ (2014) [24], where the artist controls water wave patterns with a Brain-Computer-Interface in front of an audience – or put the user into a observing position – like ‘Resonant Waves’ by Grilloti et. al (2020) [9], where the user watches the visuals of a sound and water interplay. Recently, the works of Lachlan Turczan have gained popularity, especially through their artistic work ‘Shaped by Water’ (2023) in collaboration with Google [28]. Here too, the user remains an observer. Therefore, it is our concern to place the user and the feedback into an interactive dynamic. We achieve this by augmenting the setup (vibration - water) by a respiratory input, since, to the best of our knowledge, the feedback of water waves has not been investigated in a contactless breathing interaction before. Through the use of a computer vision sensor, we are able to design a novel spatial interaction between user and natural matter. Finally, through mesmerizing and well-tempered feedback, we create a moment of self-awareness in harmony with water.

2.2 Final Design

Der Trog is a contactless breath-to-water interface that uses computer vision to track respiration. The user approaches an object filled with water, takes a seat on a chair and starts breathing: the water reacts with a wave-like movement. While the user inhales, the tension is built; while they exhale, the tension is released. A thermal computer vision system next to the object recognises small temperature differences on the user’s skin and derives the breathing rhythm and intensity without physical contact. A vibration motor underneath the water basin finally emits vibration-patterns equivalent to the breathing rhythm of the user. The object is therefore shaped by a real-time data stream from the user’s body. We aimed for a minimal, calming aesthetic, to allow the user to focus entirely on their breathing and the responsive water feedback, while minimizing visible traces of technology. At this stage of development,

the user must wear a facemask, as this enhances the respiration tracking capability of our hardware setup.

2.2.1 Hardware. The main construction is a cubic structure (1000 mm × 1000 mm × 360 mm) supporting a powder-coated sheet metal basin on its top surface, encased in black MDF. A thermal camera (384 × 288 resolution at 9 Hz) is positioned about 150 cm away, capturing subtle temperature variations around the nostrils without physical contact. Beneath the basin, a 400 W bass shaker (5–200 Hz frequency range) is secured to the underside of the metal surface, while a second bass shaker is attached to the covering for more variable wave patterns. To reduce vibration transfer to the floor, the entire construction rests on specialized spherical dampers. Inside the cube, a computer processes data from the thermal camera (connected via USB) and sends a control signal to an amplifier, which in turn powers both bass shakers. All cabling is routed away from the water to maintain safety and a clean appearance.

2.2.2 Software. Our software system is designed to translate a user’s breathing into tangible water wave patterns in real time. Implemented in TouchDesigner, the pipeline begins with a thermal camera (TE-Q1) feed. We rely on Google’s open-source MediaPipe pose tracking library [18] to detect and outline the user’s upper facial region, specifically targeting the mouth and nose area as our region of interest (ROI). In addition, we use a GPU-accelerated background removal technique to reduce ambient heat signatures and isolate the user’s thermal profile. Once the user’s facial ROI is identified, we calculate average pixel intensities on each frame to detect the moment-to-moment temperature shifts caused by inhalation (cooler air) and exhalation (warmer air). We then monitor the slope (rate of change) of these pixel values against a fluid baseline temperature, which updates continuously and allows the system to adapt to varying ambient conditions. Whenever an inhale is detected, a 15 Hz sine-wave audio signal is routed to an amplifier driving the bass shakers beneath the water basin. Exhalation yields no signal, letting existing waves persist or dissipate naturally. In extensive trials, we found that emitting a constant 15 Hz vibration during each inhalation provides a strong, coherent wave pattern on the water’s surface. Varying or shifting frequencies over time introduced overlapping oscillations and reduced amplitude, making the waveforms less distinct. By maintaining a fixed, mid-range frequency, we ensure a stable mechanical energy transfer to the fluid and minimize interference effects, allowing each inhalation to be clearly represented in the evolving wave field.

3 Conclusion

It is essential for human-computer-interaction researchers and interaction designers (and finally for the users) to recognise and address the consequences of the dominant human-centred attitude towards the environment [8]. A narrow anthropocentric perspective can range from focusing solely on the self (egoism) to encompassing all humans (personalism or humanism, including both present and future generations). This moral consideration can then be extended further to include all animals capable of suffering (pathocentrism), all living organisms (biocentrism), and ultimately all of nature (holism). Holism maintains that every element of nature possesses intrinsic value and should be protected for its own

sake [15]. Der Trog, in this constellation, investigates novel embodied and empathic access to an holistic perspective on nature and ourselves.

We aimed to design an associative, sensual experience that connects a user's conscious breathing to the natural matter of water, emphasizing their corporal connection. Der Trog investigates whether AI as an interaction tool can help create physical, sensory experiences concurrently with our body and a natural material, finally creating appreciation and compassion. Der Trog shall spark fascination for the next time the user actually encounters water in a natural environment. To relearn a way of life that is oriented towards a healthy relationship with nature and to understand it as an inherently valuable experience, one that is worth protecting simply for its existence [7, 26]. While AI may not be able to take the final, emotional step for humans towards a deeper connection with the co-environment ('Mitwelt'), technology does have the ability to create access that allows us to experience natural elements in unprecedented sensory ways, through an augmented manner.

4 Future Work

The application of AI as facilitator of interaction is only in the beginnings. With improvements in sensor technology, more trackable physiological information that can serve as input. In the future, establishing novel connections between body and nature could pave the way towards a balanced relation of humans and nature. To understand this potential, an evaluation plan could involve assessing the ability of systems to foster empathy for nature to explore emotional impact and user engagement, while tracking changes in environmental awareness and attitudes. Going further in the development of Der Trog as an installation, we aim to advance contactless breath tracking by implementing a dual-sensor system that combines thermal and colour imaging in real time. This approach enables precise tracking of the nose and mouth region without requiring a physical facemask. Leveraging a device like the OAK-T camera, which integrates thermal and RGB sensors with 4 TOPS of onboard processing power, could significantly minimise dependence on external hardware. Additionally, replacing the TouchDesigner-based workflow with a Python-only ecosystem would support a more modular and compact sensor design, simplifying data handling and unlocking opportunities for future applications.

Within the applied field of Human-Computer-Interaction, as prototype for a contactless respiration-feedback-interface, Der Trog aligns with the broader research landscape of 'Affective Matter' [23] – exploring how material environments can enhance health and wellbeing through the design of human-material interactions [4, 6]. Beyond its eco-philosophical approach to wellbeing, Der Trog has the future potential to contribute to approaches that connect embodiment and emotion processing [1, 3, 22]. For instance, it could support breathwork-based therapies [13, 19] used to treat trauma, anxiety, depression, and other mental health conditions by leveraging the interconnection between respiratory patterns and emotion regulation [12, 14, 29].

Every environment or object has the potential to influence our mood and emotions through its tactile, visual, acoustic qualities, and cultural significance. As designers, we hold the ability to craft

the sensory and emotional characteristics of spaces, objects, and the interactions they enable. In light of emerging technological advancements, the augmentation of conventional physical interaction – and the resulting altered perception of human corporality within its environmental context – constitutes an increasingly significant area of research. Der Trog represents a materialized example of a broader possible class of technologically-enabled human-environment interfaces, and serves as a conceptual blueprint for reimagining the use of the human body in novel interactions with our surroundings.

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